

## Run: alpha-ramp-150e

## Log-log loss curves

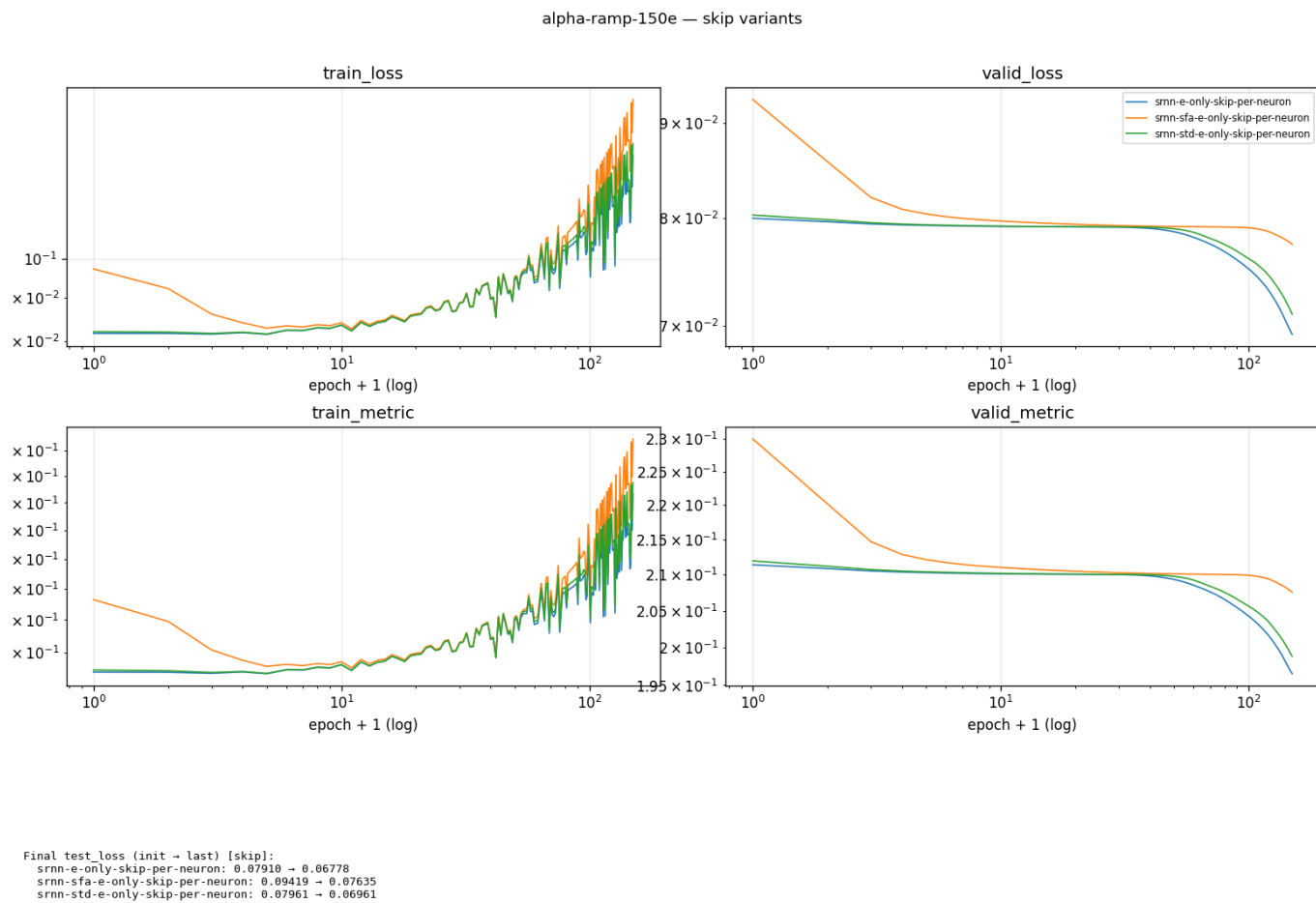


Figure 1: skip variants — log-log loss/metric

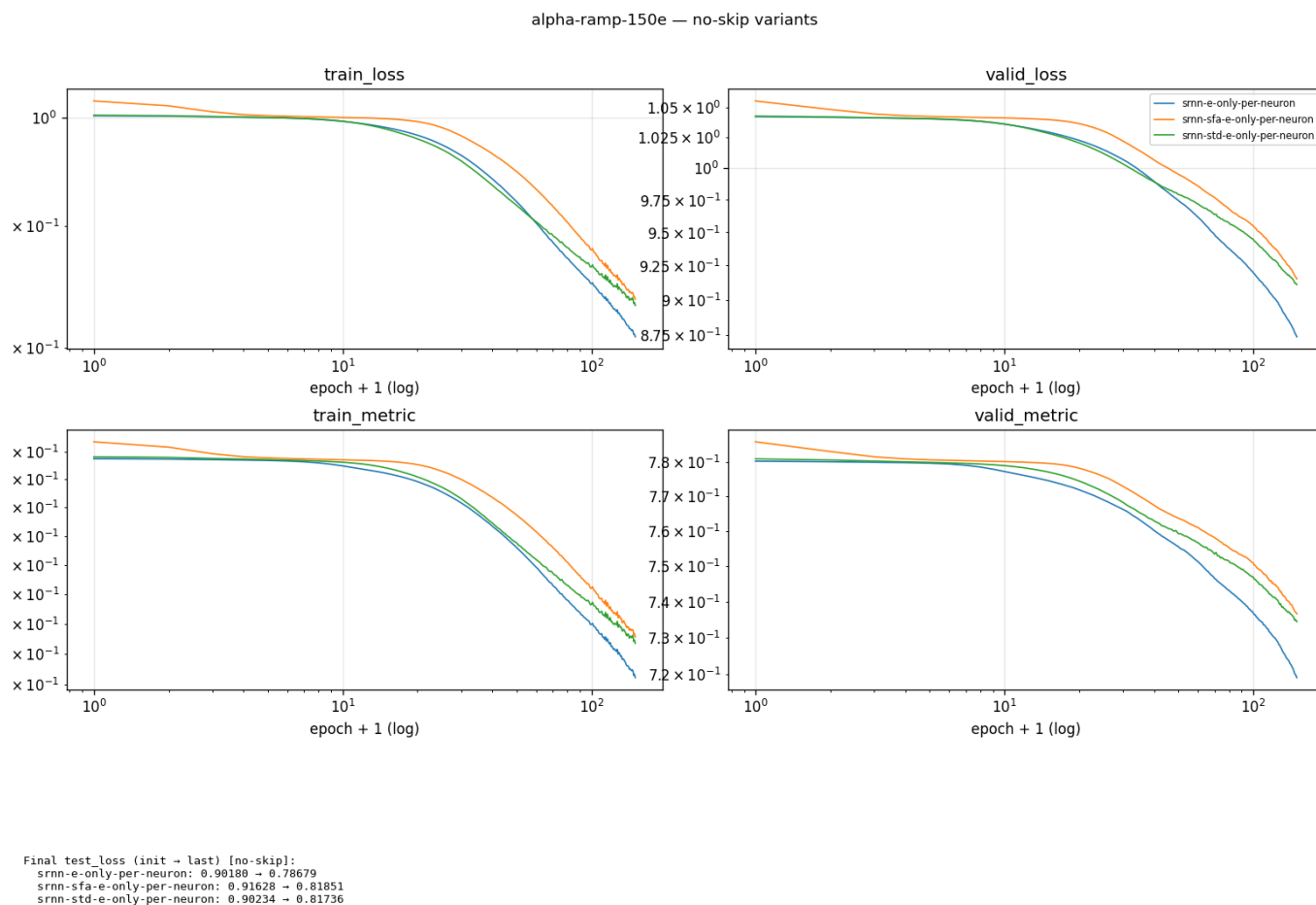


Figure 2: no-skip variants — log-log loss/metric

## srnn-e-only-per-neuron

alpha-ramp-150e — effective time constants (srnn-e-only-per-neuron)

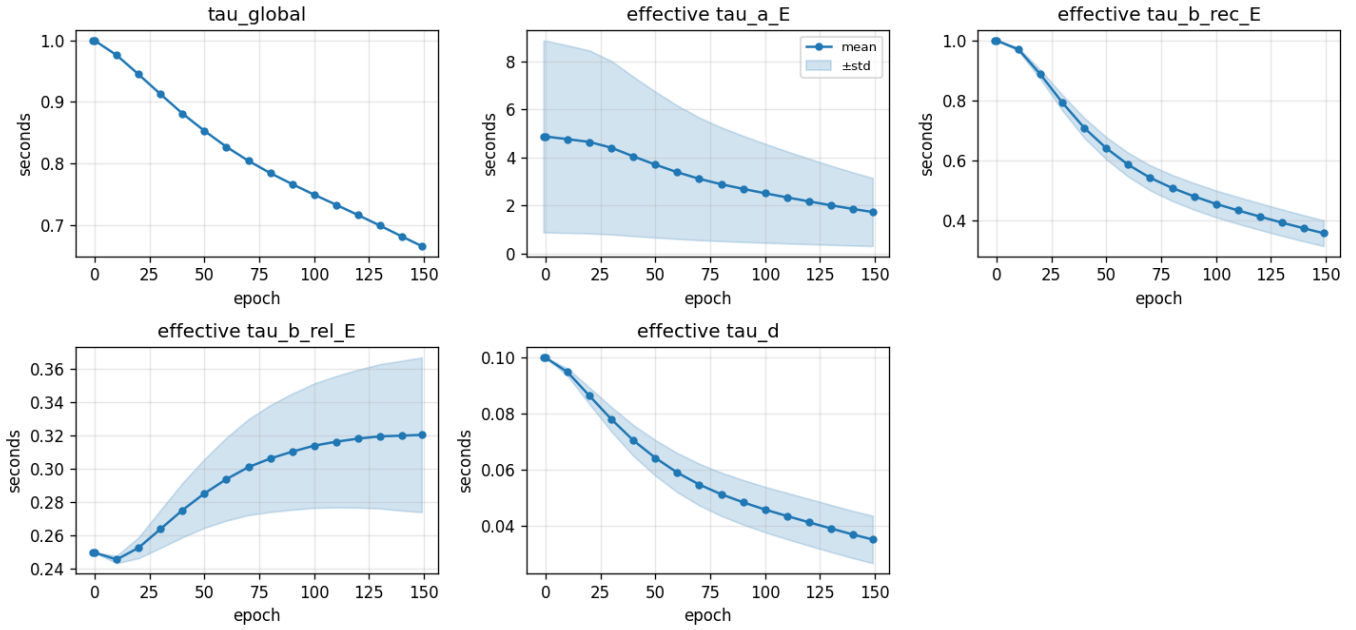
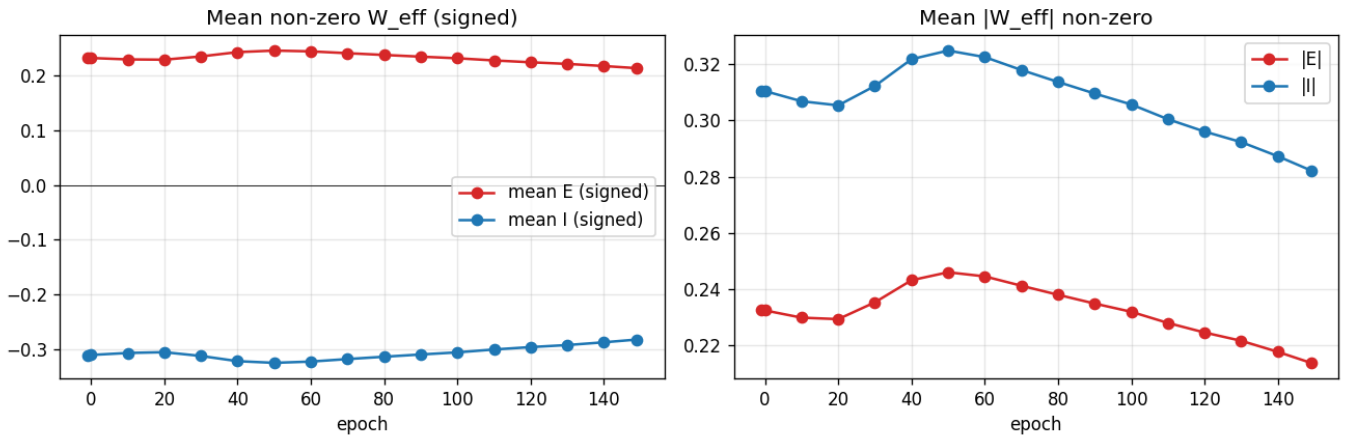


Figure 3: Tau evolution

alpha-ramp-150e — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4:  $W_{\text{EI}}$  evolution

## srnn-e-only-per-neuron — parameter table

# alpha-ramp-150e - variant srnn-e-only-per-neuron (k=0)  
# Effective values (post-transform). Means/stds computed over the indicated dimension.

| effective parameter      | init mean   | final mean | init std | final std | n      |
|--------------------------|-------------|------------|----------|-----------|--------|
| -----                    |             |            |          |           |        |
| tau_global (s)           | +1          | +0.66543   | -        | -         | scalar |
| tau_d (s)                | +0.1        | +0.035117  | 0        | 0.008453  | 300    |
| tau_a_E (s)              | +4.8777     | +1.7311    | 3.996    | 1.418     | 450    |
| tau_b_rec_E (s)          | +1          | +0.35754   | 0        | 0.04294   | 150    |
| tau_b_rel_E (s)          | +0.25       | +0.32035   | 0        | 0.04641   | 150    |
| W_eff (E src, signed)    | +0.23267    | +0.21389   | 0.07641  | 0.07508   | 15048  |
| W_eff (I src, signed)    | -0.31062    | -0.28226   | 0.07724  | 0.07799   | 14977  |
| W_eff  (E src)           | +0.23267    | +0.21389   | 0.07641  | 0.07508   | 15048  |
| W_eff  (I src)           | +0.31062    | +0.28226   | 0.07724  | 0.07799   | 14977  |
| W_in_eff (input neurons) | -0.0003386  | +0.001217  | 0.1012   | 0.1355    | 6675   |
| c_E (SFA coupling)       | +0.05       | +0.06368   | 0        | 0.01052   | 450    |
| c_0_E (SFA offset)       | +0          | -0.28228   | 0        | 0.1187    | 450    |
| a_0 (threshold)          | +0.35       | +0.17353   | 0        | 0.09971   | 300    |
| readout_weight           | -2.8854e-05 | -0.0041653 | 0.06604  | 0.1689    | 6675   |
| readout_bias             | +0          | -0.0014186 | 0        | 0.1764    | 89     |
| ic.ic (per-variant init) | +0.098875   | -0.010233  | 0.3842   | 0.5166    | 900    |

## srnn-e-only-skip-per-neuron

alpha-ramp-150e — effective time constants (srnn-e-only-skip-per-neuron)

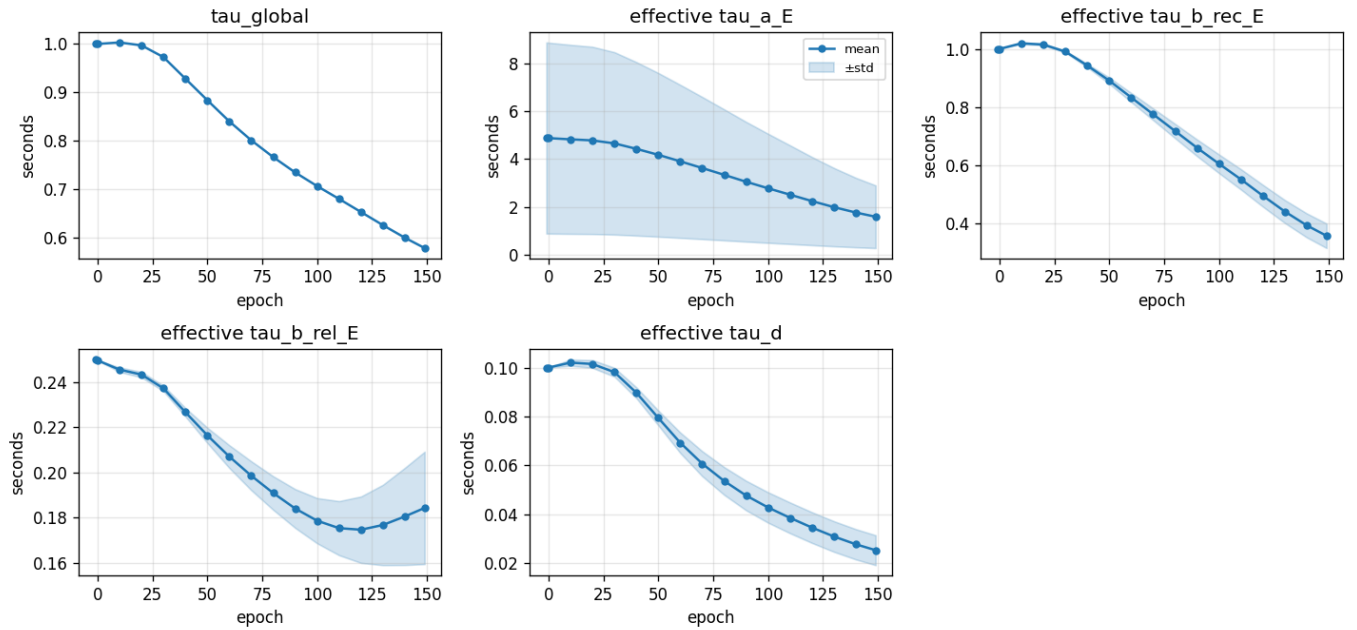
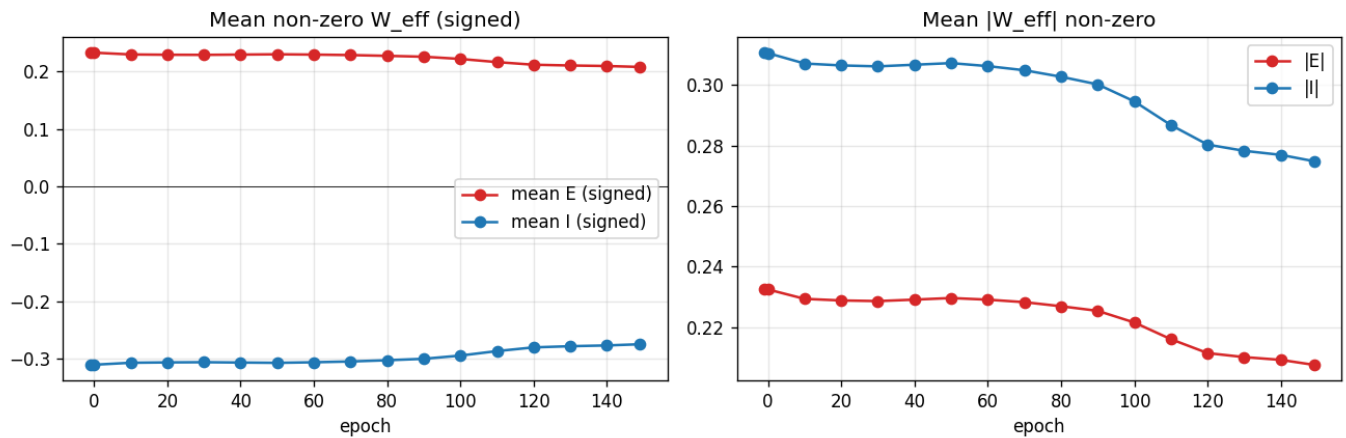


Figure 5: Tau evolution

alpha-ramp-150e — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6:  $W_{EI}$  evolution

## srnn-e-only-skip-per-neuron — parameter table

# alpha-ramp-150e - variant srnn-e-only-skip-per-neuron (k=1)  
# Effective values (post-transform). Means/stds computed over the indicated dimension.

| effective parameter      | init mean   | final mean  | init std | final std | n      |
|--------------------------|-------------|-------------|----------|-----------|--------|
| tau_global (s)           | +1          | +0.57849    | -        | -         | scalar |
| tau_d (s)                | +0.1        | +0.025282   | 0        | 0.006126  | 300    |
| tau_a_E (s)              | +4.8777     | +1.585      | 3.996    | 1.31      | 450    |
| tau_b_rec_E (s)          | +1          | +0.35732    | 0        | 0.04221   | 150    |
| tau_b_rel_E (s)          | +0.25       | +0.18424    | 0        | 0.02491   | 150    |
| W_eff (E src, signed)    | +0.23267    | +0.20765    | 0.07641  | 0.07126   | 15048  |
| W_eff (I src, signed)    | -0.31062    | -0.27481    | 0.07724  | 0.07415   | 14977  |
| W_eff  (E src)           | +0.23267    | +0.20765    | 0.07641  | 0.07126   | 15048  |
| W_eff  (I src)           | +0.31062    | +0.27481    | 0.07724  | 0.07415   | 14977  |
| W_in_eff (input neurons) | -0.00084237 | -0.00087285 | 0.1008   | 0.04768   | 6675   |
| c_E (SFA coupling)       | +0.05       | +0.076048   | 0        | 0.01365   | 450    |
| c_0_E (SFA offset)       | +0          | -0.20905    | 0        | 0.1452    | 450    |
| a_0 (threshold)          | +0.35       | +0.14821    | 0        | 0.1304    | 300    |
| readout_weight           | +0.00078219 | -0.0072867  | 0.06654  | 0.109     | 6675   |
| readout_bias             | +0          | +0.11416    | 0        | 0.1458    | 89     |
| ic.ic (per-variant init) | +0.098961   | -0.033766   | 0.387    | 0.4815    | 900    |

## srnn-sfa-e-only-per-neuron

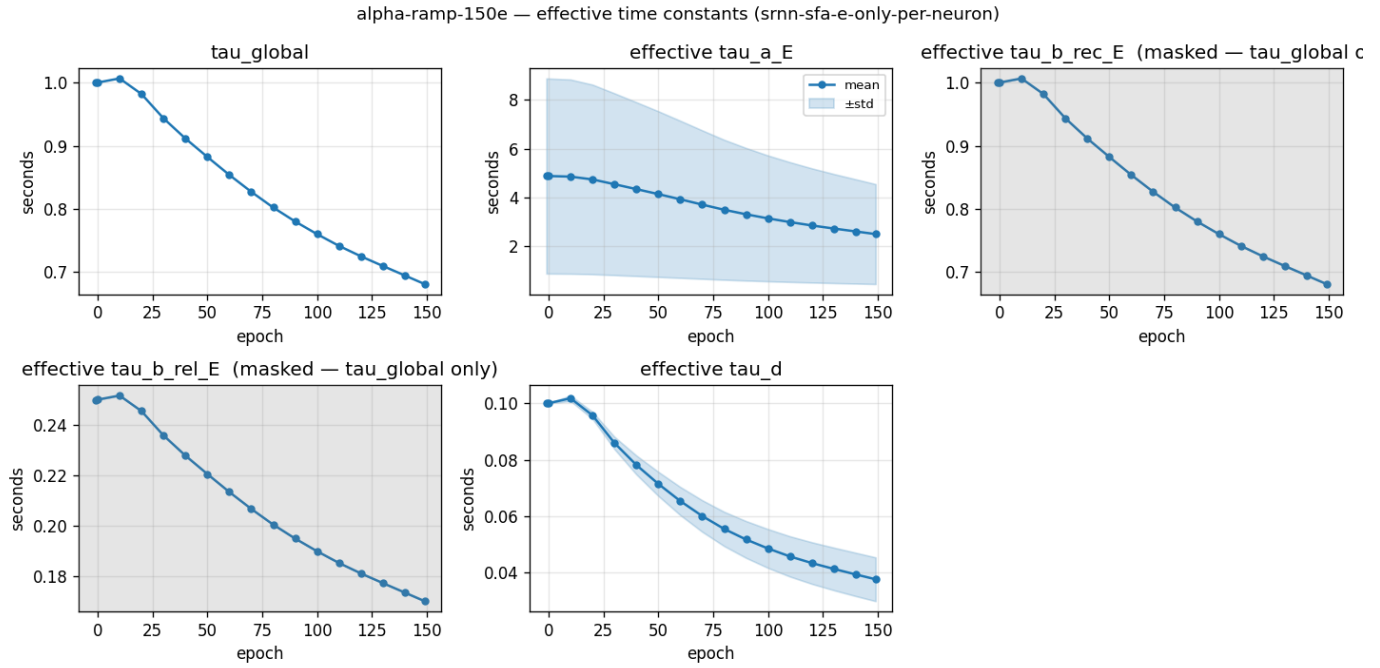
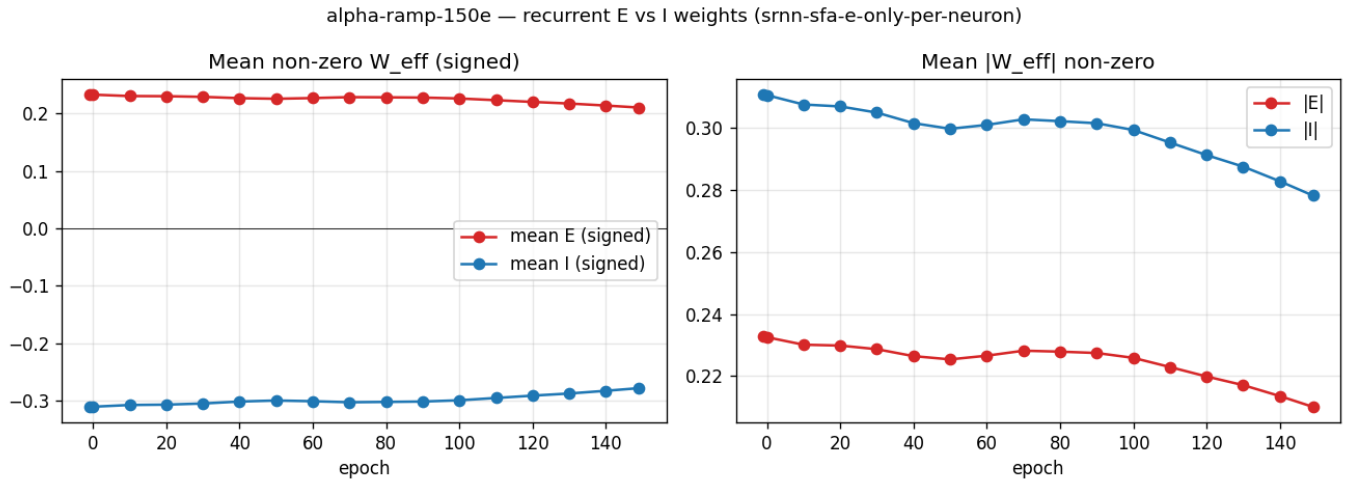


Figure 7: Tau evolution

Figure 8:  $W_{EI}$  evolution

## srnn-sfa-e-only-per-neuron — parameter table

# alpha-ramp-150e - variant srnn-sfa-e-only-per-neuron (k=2)  
# Effective values (post-transform). Means/stds computed over the indicated dimension.

| effective parameter      | init mean   | final mean | init std | final std | n      |
|--------------------------|-------------|------------|----------|-----------|--------|
| tau_global (s)           | +1          | +0.6809    | -        | -         | scalar |
| tau_d (s)                | +0.1        | +0.037426  | 0        | 0.007794  | 300    |
| tau_a_E (s)              | +4.8777     | +2.4981    | 3.996    | 2.047     | 450    |
| tau_b_rec_E (s)          | +1          | +0.6809    | 0        | 0         | 150    |
| tau_b_rel_E (s)          | +0.25       | +0.17023   | 0        | 0         | 150    |
| W_eff (E src, signed)    | +0.23267    | +0.21012   | 0.07641  | 0.07208   | 15048  |
| W_eff (I src, signed)    | -0.31062    | -0.27813   | 0.07724  | 0.07452   | 14977  |
| W_eff  (E src)           | +0.23267    | +0.21012   | 0.07641  | 0.07208   | 15048  |
| W_eff  (I src)           | +0.31062    | +0.27813   | 0.07724  | 0.07452   | 14977  |
| W_in_eff (input neurons) | -0.00015382 | -0.0021955 | 0.09934  | 0.1421    | 6675   |
| c_E (SFA coupling)       | +0.05       | +0.051488  | 0        | 0.005558  | 450    |
| c_0_E (SFA offset)       | +0          | -0.17221   | 0        | 0.103     | 450    |
| a_0 (threshold)          | +0.35       | +0.35374   | 0        | 0.06641   | 300    |
| readout_weight           | -0.00098972 | +0.0074873 | 0.06678  | 0.1426    | 6675   |
| readout_bias             | +0          | -0.047463  | 0        | 0.08236   | 89     |
| ic.ic (per-variant init) | +0.13078    | +0.0259    | 0.4172   | 0.4526    | 900    |

## srnn-sfa-e-only-skip-per-neuron

alpha-ramp-150e — effective time constants (srnn-sfa-e-only-skip-per-neuron)

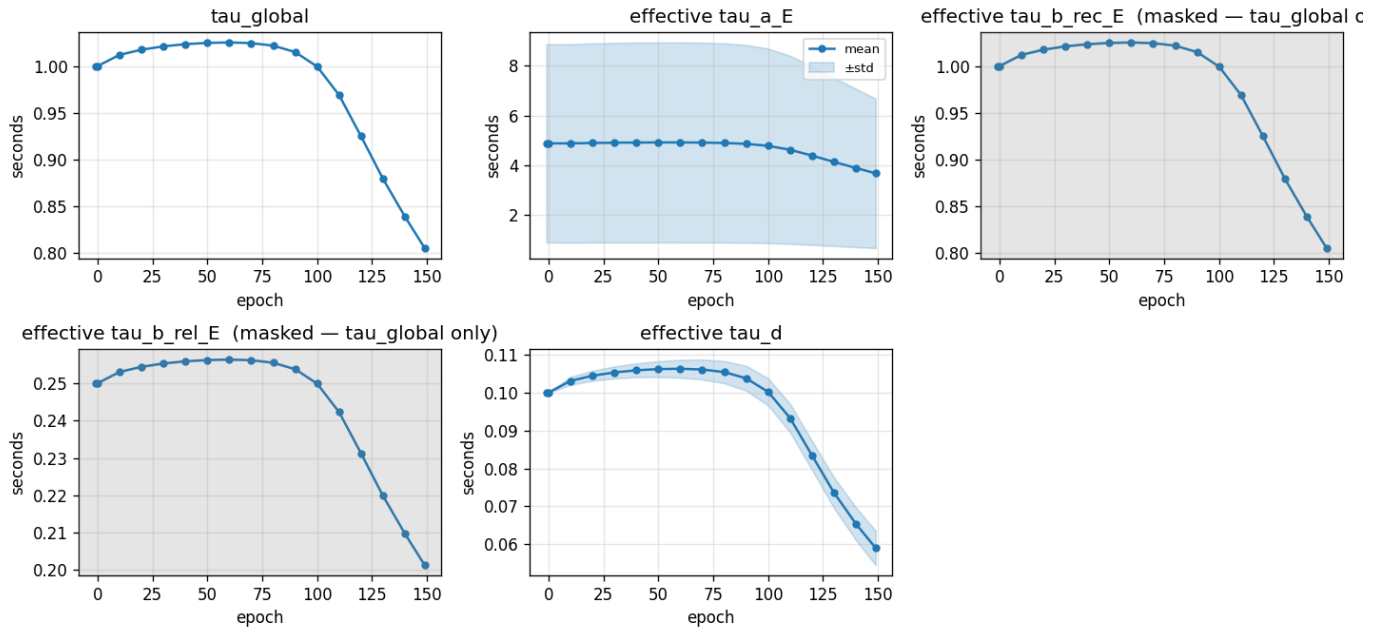
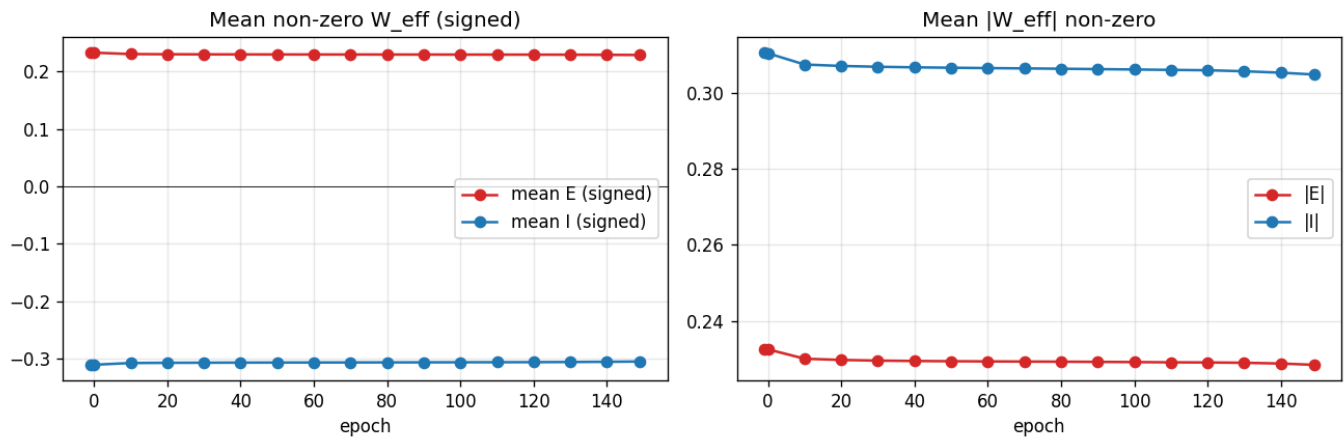


Figure 9: Tau evolution

alpha-ramp-150e — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10:  $W_{EI}$  evolution

## srnn-sfa-e-only-skip-per-neuron — parameter table

# alpha-ramp-150e - variant srnn-sfa-e-only-skip-per-neuron (k=3)  
# Effective values (post-transform). Means/stds computed over the indicated dimension.

| effective parameter      | init mean   | final mean | init std | final std | n      |
|--------------------------|-------------|------------|----------|-----------|--------|
| tau_global (s)           | +1          | +0.80508   | -        | -         | scalar |
| tau_d (s)                | +0.1        | +0.058983  | 0        | 0.004545  | 300    |
| tau_a_E (s)              | +4.8777     | +3.668     | 3.996    | 3.011     | 450    |
| tau_b_rec_E (s)          | +1          | +0.80508   | 0        | 0         | 150    |
| tau_b_rel_E (s)          | +0.25       | +0.20127   | 0        | 0         | 150    |
| W_eff (E src, signed)    | +0.23267    | +0.22845   | 0.07641  | 0.07569   | 15048  |
| W_eff (I src, signed)    | -0.31062    | -0.30485   | 0.07724  | 0.07676   | 14977  |
| W_eff  (E src)           | +0.23267    | +0.22845   | 0.07641  | 0.07569   | 15048  |
| W_eff  (I src)           | +0.31062    | +0.30485   | 0.07724  | 0.07676   | 14977  |
| W_in_eff (input neurons) | -0.0011621  | -0.0031755 | 0.09962  | 0.08641   | 6675   |
| c_E (SFA coupling)       | +0.05       | +0.052814  | 0        | 0.004382  | 450    |
| c_0_E (SFA offset)       | +0          | +0.02968   | 0        | 0.06652   | 450    |
| a_0 (threshold)          | +0.35       | +0.36487   | 0        | 0.06577   | 300    |
| readout_weight           | +0.00016801 | -0.0029894 | 0.0673   | 0.0635    | 6675   |
| readout_bias             | +0          | +0.032153  | 0        | 0.04143   | 89     |
| ic.ic (per-variant init) | +0.09231    | +0.090339  | 0.5294   | 0.4444    | 900    |

## srnn-std-e-only-per-neuron

alpha-ramp-150e — effective time constants (srnn-std-e-only-per-neuron)

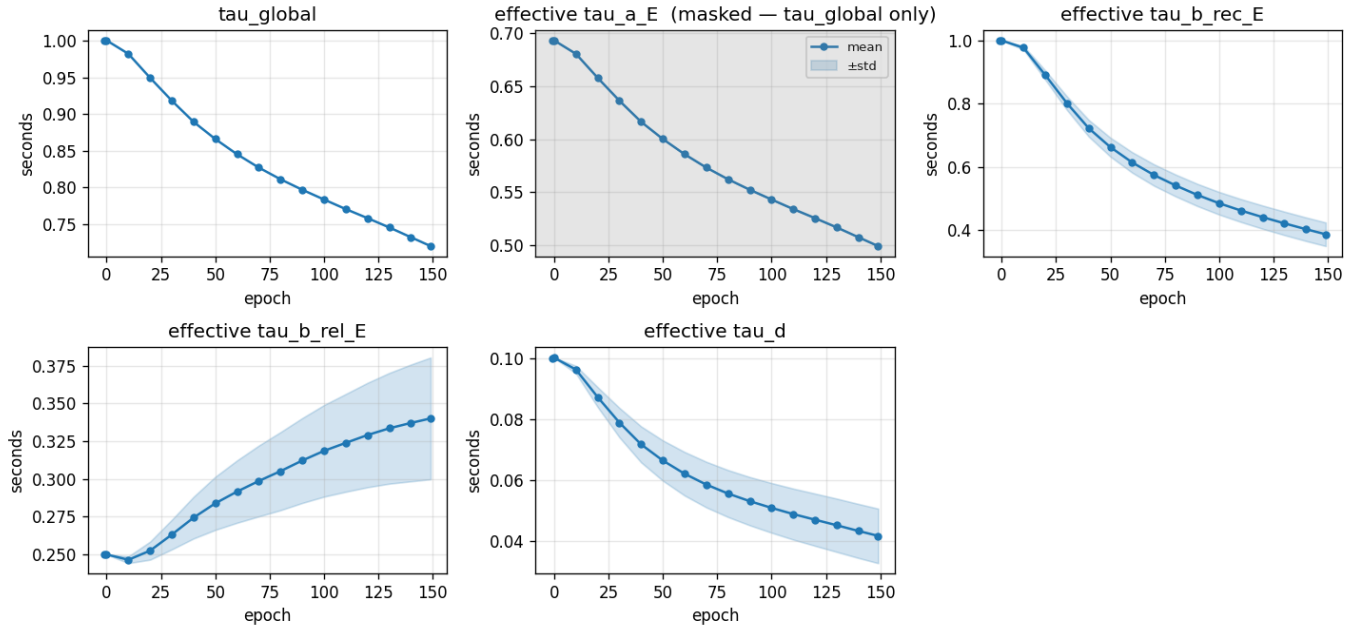
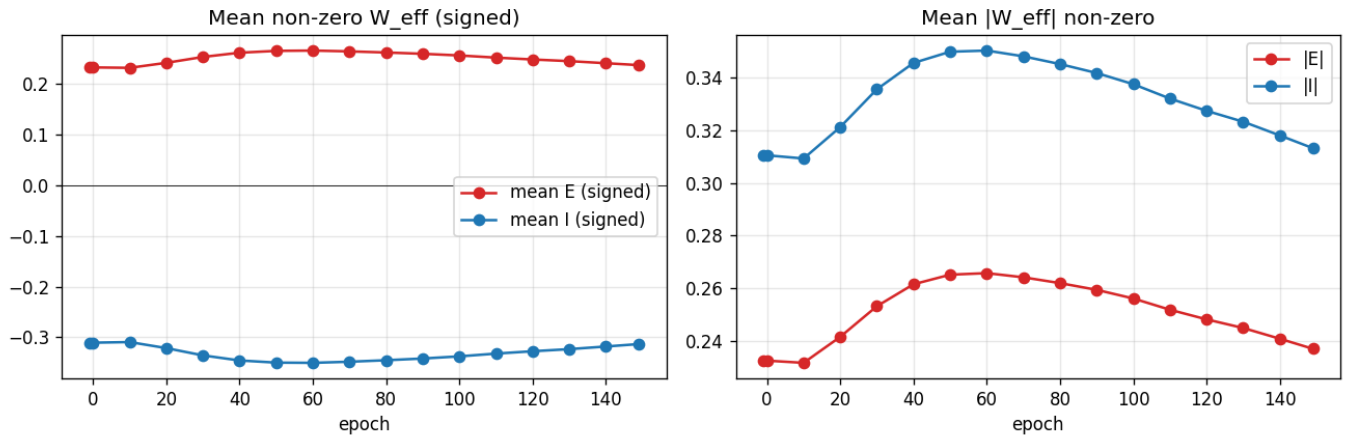


Figure 11: Tau evolution

alpha-ramp-150e — recurrent E vs I weights (srnn-std-e-only-per-neuron)

Figure 12:  $W_{\text{EI}}$  evolution

## srnn-std-e-only-per-neuron — parameter table

# alpha-ramp-150e - variant srnn-std-e-only-per-neuron (k=4)  
# Effective values (post-transform). Means/stds computed over the indicated dimension.

| effective parameter      | init mean   | final mean | init std | final std | n      |
|--------------------------|-------------|------------|----------|-----------|--------|
| tau_global (s)           | +1          | +0.72008   | -        | -         | scalar |
| tau_d (s)                | +0.1        | +0.041684  | 0        | 0.008946  | 300    |
| tau_a_E (s)              | +0.69315    | +0.49912   | 0        | 0         | 450    |
| tau_b_rec_E (s)          | +1          | +0.38513   | 0        | 0.0375    | 150    |
| tau_b_rel_E (s)          | +0.25       | +0.34002   | 0        | 0.04036   | 150    |
| W_eff (E src, signed)    | +0.23267    | +0.23701   | 0.07641  | 0.08288   | 15048  |
| W_eff (I src, signed)    | -0.31062    | -0.31315   | 0.07724  | 0.08562   | 14977  |
| W_eff  (E src)           | +0.23267    | +0.23701   | 0.07641  | 0.08288   | 15048  |
| W_eff  (I src)           | +0.31062    | +0.31315   | 0.07724  | 0.08562   | 14977  |
| W_in_eff (input neurons) | +0.00019944 | -0.0017695 | 0.09994  | 0.1438    | 6675   |
| c_E (SFA coupling)       | +0.05       | +0.05      | 0        | 0         | 450    |
| c_0_E (SFA offset)       | +0          | +0         | 0        | 0         | 450    |
| a_0 (threshold)          | +0.35       | +0.23747   | 0        | 0.08906   | 300    |
| readout_weight           | +0.00074661 | +0.008098  | 0.06657  | 0.1618    | 6675   |
| readout_bias             | +0          | -0.029767  | 0        | 0.1688    | 89     |
| ic.ic (per-variant init) | +0.087739   | +0.035925  | 0.3868   | 0.4518    | 900    |

## srnn-std-e-only-skip-per-neuron

alpha-ramp-150e — effective time constants (srnn-std-e-only-skip-per-neuron)

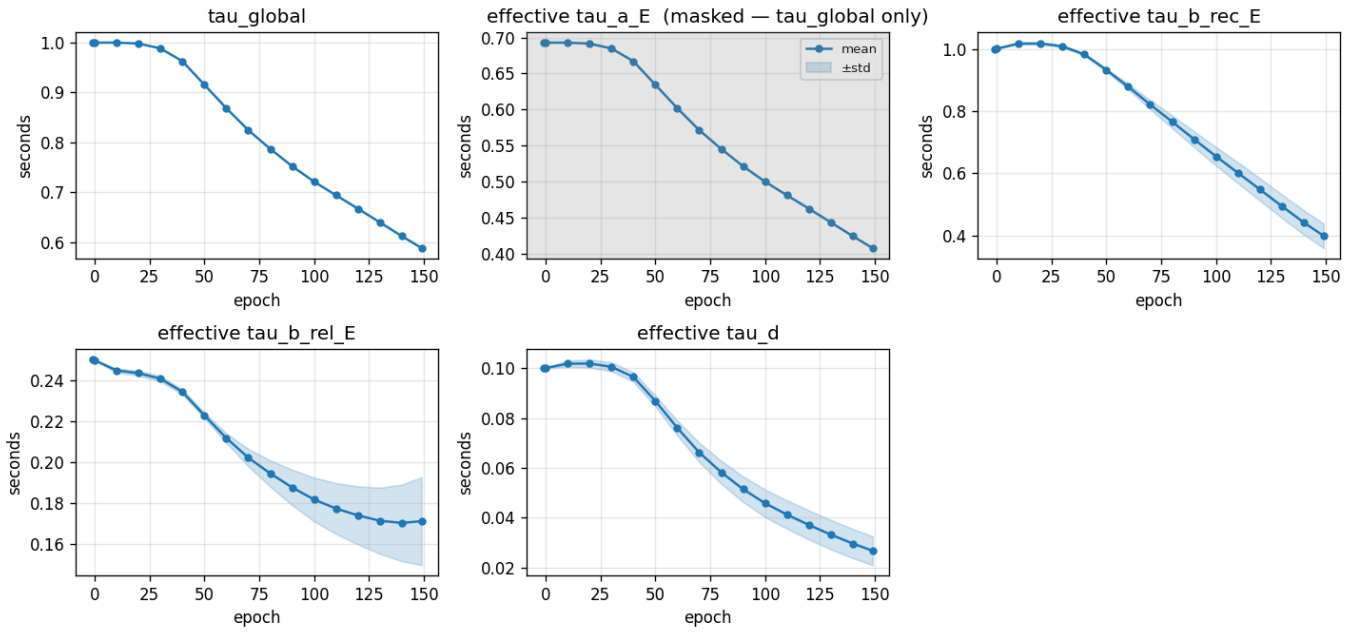
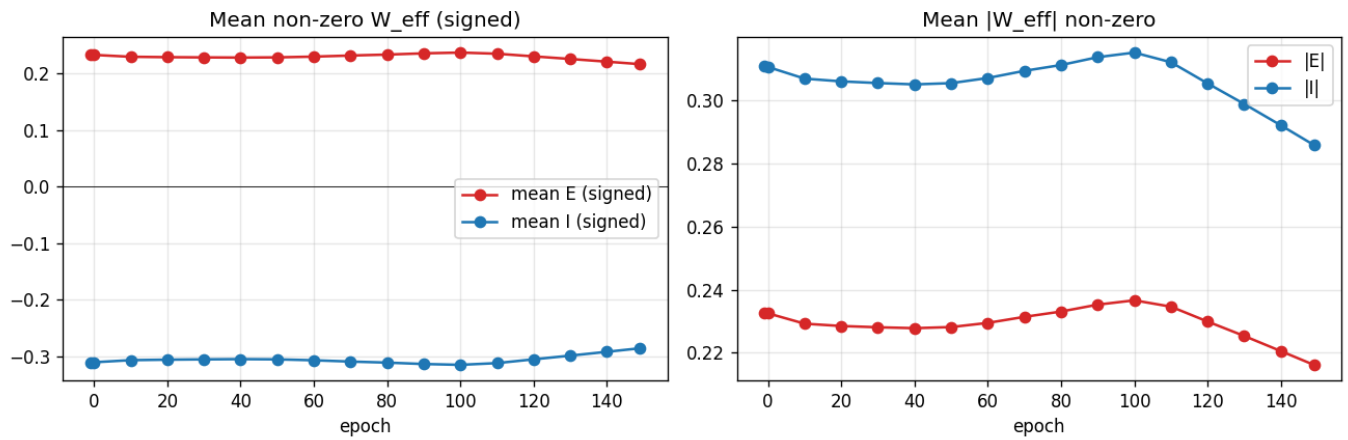


Figure 13: Tau evolution

alpha-ramp-150e — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

Figure 14:  $W_{EI}$  evolution

## srnn-std-e-only-skip-per-neuron — parameter table

# alpha-ramp-150e - variant srnn-std-e-only-skip-per-neuron (k=5)  
# Effective values (post-transform). Means/stds computed over the indicated dimension.

| effective parameter      | init mean   | final mean  | init std | final std | n      |
|--------------------------|-------------|-------------|----------|-----------|--------|
| -----                    |             |             |          |           |        |
| tau_global (s)           | +1          | +0.58789    | -        | -         | scalar |
| tau_d (s)                | +0.1        | +0.02679    | 0        | 0.005781  | 300    |
| tau_a_E (s)              | +0.69315    | +0.40749    | 0        | 0         | 450    |
| tau_b_rec_E (s)          | +1          | +0.39887    | 0        | 0.03951   | 150    |
| tau_b_rel_E (s)          | +0.25       | +0.17126    | 0        | 0.02144   | 150    |
| W_eff (E src, signed)    | +0.23267    | +0.21621    | 0.07641  | 0.07365   | 15048  |
| W_eff (I src, signed)    | -0.31062    | -0.28576    | 0.07724  | 0.07655   | 14977  |
| W_eff  (E src)           | +0.23267    | +0.21621    | 0.07641  | 0.07365   | 15048  |
| W_eff  (I src)           | +0.31062    | +0.28576    | 0.07724  | 0.07655   | 14977  |
| W_in_eff (input neurons) | +0.00036119 | +0.00024403 | 0.09956  | 0.0587    | 6675   |
| c_E (SFA coupling)       | +0.05       | +0.05       | 0        | 0         | 450    |
| c_0_E (SFA offset)       | +0          | +0          | 0        | 0         | 450    |
| a_0 (threshold)          | +0.35       | +0.19603    | 0        | 0.1144    | 300    |
| readout_weight           | +0.00055372 | -0.012366   | 0.06572  | 0.09322   | 6675   |
| readout_bias             | +0          | +0.090316   | 0        | 0.1209    | 89     |
| ic.ic (per-variant init) | +0.089141   | +0.013634   | 0.3857   | 0.4678    | 900    |